ELG 3125 - Signals & Systems

Lecture 2 - Part 3

General form of Complex Exponential Signals

Suppose we have: Cd^n, $C = |c|e^{j\theta}$ and $d = |d|e^{j\omega_0}$

$$Cd^n = |c|e^{j\theta}\left(|d|e^{j\omega_0}\right)^n = |c|e^{j\theta}\,|d|^n e^{j\omega_0 n} = |c||d|^n e^{j(\omega_0 n+\theta)}$$

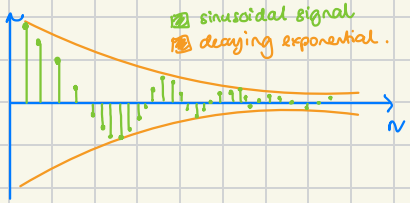
$$e^{j(\omega_0 n+\theta)} = \cos(\omega_0 n+\theta) + j\sin(\omega_0 n+\theta)$$

$$\Rightarrow Cd^n = |c||d|^n\left[\cos(\omega_0 n+\theta) + j\sin(\omega_0 n+\theta)\right] = |c||d|^n\cos(\omega_0 n+\theta) + j|c||d|^n\sin(\omega_0 n+\theta)$$

If: $|d| = 1 \Rightarrow Cd^n = \underbrace{|c|\cos(\omega_0 n+\theta)}_{\text{Real part is sinusoidal}} + j\underbrace{(|c|\sin(\omega_0 n+\theta))}_{\text{Imaginary part is sinusoidal.}}$

If: $|d| < 1 \Rightarrow Cd^n = |c||d|^n\cos(\omega_0 n+\theta) + j|c||d|^n\sin(\omega_0 n+\theta)$ $\Rightarrow$ sinusoidal signals constrained by a decaying exponential

($|d|^n$: decaying exponential)



If: $|d| > 1 \Rightarrow Cd^n = |c||d|^n\cos(\omega_0 n+\theta) + j|c||d|^n\sin(\omega_0 n+\theta)$ $\Rightarrow$ sinusoidal signals constrained by a growing exponential.

($|d|^n$: growing exponential)



NOTE: When dealing with discrete-time exponentials $\Rightarrow$ only consider a frequency interval of 2π

use intervals: $0 \le \omega_0 \le 2\pi$ or $-\pi \le \omega_0 \le \pi$

What is the condition for $x[n] = e^{j\omega_0 n}$ to be periodic with period $N > 0$?

$x[n] = x[n+N]$

$x[n+N] = e^{j\omega_0(n+N)} = e^{j\omega_0 n}e^{j\omega_0 N} = e^{j\omega_0 n}$

$\Rightarrow e^{j\omega_0 N} = \dfrac{e^{j\omega_0 n}}{e^{j\omega_0 n}} = 1 \Rightarrow e^{j\omega_0 N} = 1 = \cos(\omega_0 N) + j\sin(\omega_0 N) \Rightarrow \cos(\omega_0 N) = 1 \Rightarrow \omega_0 N = m(2\pi)$, $m \in \mathbb{Z}$ (a multiple of 2π)

$\Rightarrow$ Condition: $\boxed{\dfrac{\omega_0}{2\pi} = \dfrac{m}{N}}$ Fundamental period of the signal: $\boxed{N = m\left(\dfrac{2\pi}{\omega_0}\right)}$ (N must be an integer)

Example 1: Determine fundamental period of the discrete-time signal: $x[n] = e^{j(2\pi/3)n} + e^{j(3\pi/4)n}$

$x[n] = e^{j(2\pi/3)n} + e^{j(3\pi/4)n} = x_1[n] + x_2[n]$

$\Rightarrow x_1[n] = e^{j(2\pi/3)n}$, $\omega_0 = 2\pi/3$

$N_1 = m\left(\frac{2\pi}{\omega_0}\right) = m\left(\frac{2\pi}{(2\pi/3)}\right) = 3m = 3$, $m = 1$ $\Rightarrow$ $\boxed{N_1 = 3}$

$\Rightarrow x_2[n] = e^{j(3\pi/4)n}$, $\omega_0 = 3\pi/4$

$N_2 = m\left(\frac{2\pi}{\omega_0}\right) = m\left(\frac{2\pi}{(3\pi/4)}\right) = m\left(\frac{8}{3}\right) = 8$, $m = 3$ $\Rightarrow$ $\boxed{N_2 = 8}$

Find LCM (least common multiple) of N_1 and N_2

Suppose you have 2 numbers: a, b

$\Rightarrow LCM(a,b) = \frac{(a)(b)}{GCD(a,b)}$

$GCD(a,b)$: Greatest common divisor of a, b.

$GCD(3,8) = 1$

$LCM(3,8) = \frac{(3)(8)}{1} = 24 \Rightarrow \boxed{N = 24}$ fundamental period.

Example 2: Is $x[n] = e^{j(2/3)n}$, $\omega_0 = 2/3$ periodic? If yes $\Rightarrow$ find period. If No $\Rightarrow$ explain why.

$N = m\left(\frac{2\pi}{\omega_0}\right) = m\left(\frac{2\pi}{(2/3)}\right) = 3\pi(m)$, there is no possible integer value of m that makes N an integer.

$\Rightarrow$ signal is not periodic.

Example 3: Is $x[n] = e^{j(3\pi/5)n} + e^{j(5\pi/7)n}$ periodic? If yes $\Rightarrow$ find period. If No $\Rightarrow$ explain why.

$x[n] = e^{j(3\pi/5)n} + e^{j(5\pi/7)n} = x_1[n] + x_2[n]$

$x_1[n] = e^{j(3\pi/5)n}$, $\omega_0 = 3\pi/5$

$x_2[n] = e^{j(5\pi/7)n}$, $\omega_0 = 5\pi/7$

$N_1 = m\left(\frac{2\pi}{(3\pi/5)}\right) = m\left(\frac{10}{3}\right) = 10$, $m = 3$ $\Rightarrow$ $\boxed{N_1 = 10}$

$N_2 = m\left(\frac{2\pi}{(5\pi/7)}\right) = m\left(\frac{14}{5}\right) = 14$, $m = 5$ $\Rightarrow$ $\boxed{N_2 = 14}$

$GCD(10,14) = 2$

$LCM(10,14) = \frac{(10)(14)}{2} = 70 \Rightarrow \boxed{N = 70}$ fundamental period.

Harmonically related periodic exponentials

$\varphi_k[n] = e^{jk(2\pi/N)n}$, $k = 0, \pm 1, \ldots$

$\varphi_{k+N}[n] = e^{j(k+N)(2\pi/N)n} = e^{jk(\frac{2\pi}{N})n} e^{jN(\frac{2\pi}{N})n} = e^{jk(\frac{2\pi}{N})n} e^{jn(2\pi)} = e^{jk(2\pi/N)n} = \varphi_k[n] \Rightarrow \boxed{\varphi_{k+N}[n] = \varphi_k[n]}$

(n integer; $e^{jn(2\pi)} = \cos(n(2\pi)) + j\sin(n(2\pi)) = 1$)